



inClass- L4

CV2090

Probability & Statistics for Civil Engineers

(Jul – Nov 2026)

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Flow

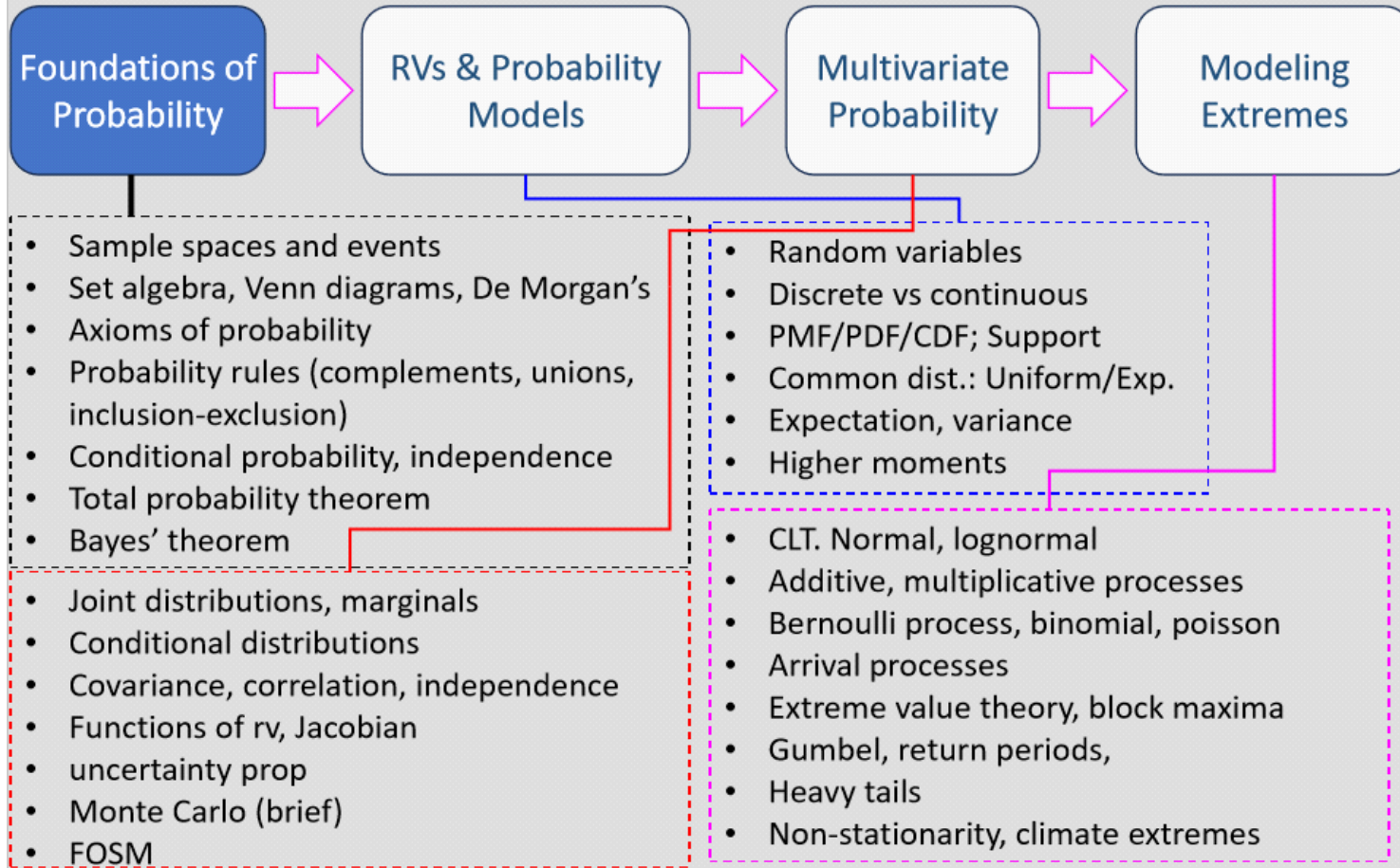
- Announcement
- Recap
- Module-1 continues

Announcement

- First **tutorial today** (3rd August; **3–5 pm**) → @ 4:15 pm
 - Venue: **BSB 203** (same as the classroom) *Announcement of Qs to be submitted.*
- Check Moodle regularly (courses.iitm.ac.in)
 - *Uploaded (Courses → CV2090 → General → Batch-2 notes)*
 - *the course outline*
 - *In-class notes from the first three lectures*

Module I: Chapter 1 (M1C1)

Module 1: **Probability** – Understanding Uncertainty



Messages from the Last Class

- **Message 1:** Biases/intuition

- Birthday Problem: Intuition can deceive you
- Reganoh/Linda: Representativeness Heuristic / Stereotyping / familiar story
- Boy-Girl problem: Equiprobability bias

- **Message 2:** Bertrand Paradox:

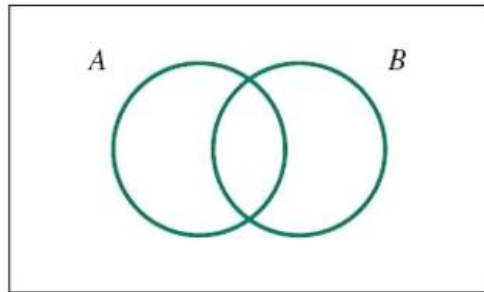
- Different randomization may lead to distinct Prob.

- **Message 3:** Conventional probability definition fails

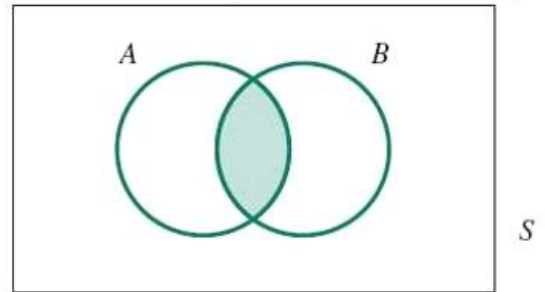
Axiomatic Approach

- Non-negativity $P(A) \geq 0$
- Normalization $P(S) = 1$
- Additivity, if $A \cap B = \emptyset$, $P(A \cup B) = P(A) + P(B)$

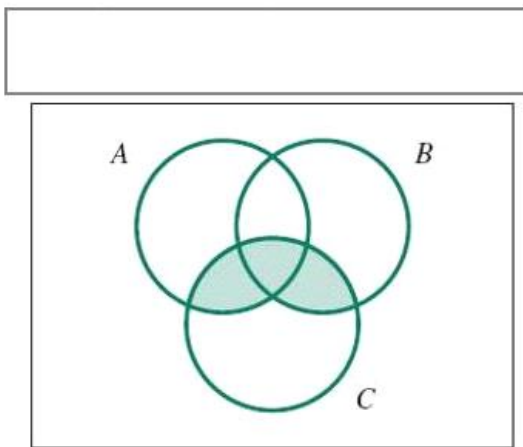
Venn Diagrams



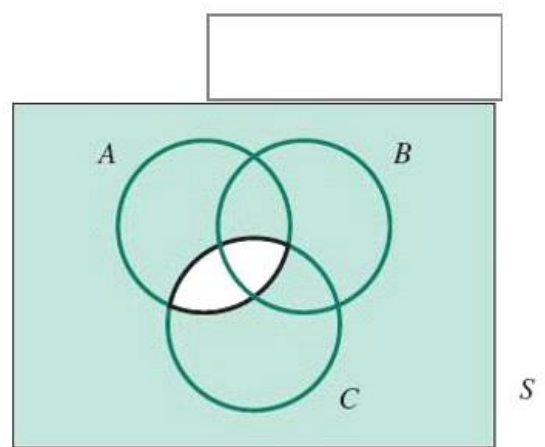
(a)



(b)



(c)



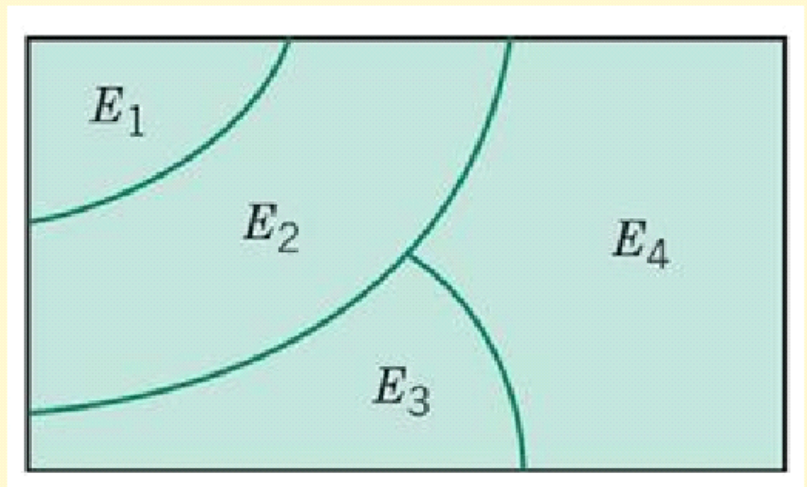
(d)

Set Theory: Mutually Exclusive Events

If $E_1, E_2, \dots, E_n$ are such that

for all pairs, $E_i \cap E_j = \emptyset \leftarrow$ Mutually Exclusive

$$Pr(E_1 \cup E_2 \cup \dots \cup E_n) = Pr(E_1) + Pr(E_2) + \dots + Pr(E_n)$$



Set Theory: MECE

Mutually Exclusive Collectively Exhaustive

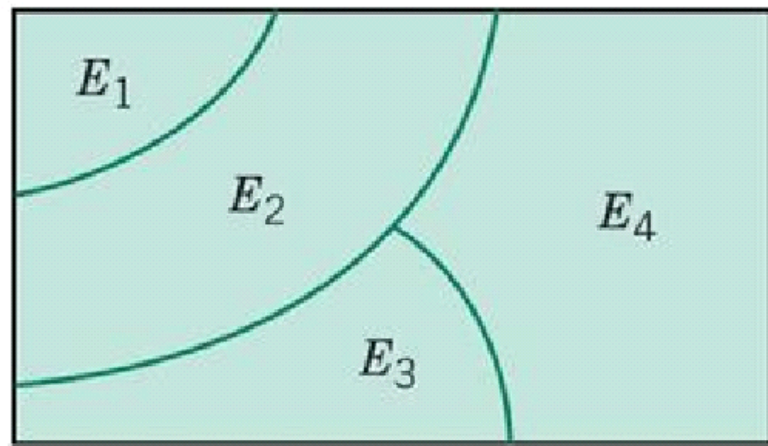
$$E_i \cap E_j = \emptyset \quad \forall i, j$$

$$\bigcup_{j=1}^n E_j = S \quad \leftarrow \text{collectively exhaustive}$$

Example:

→ Branches

→ Age groups - App



Set Theory: Other Rules

- Commutative rule

$$E_1 \cup E_2 = E_2 \cup E_1$$

$$E_1 \cap E_2 = E_2 \cap E_1$$

- Associative rule

$$(E_1 \cup E_2) \cup E_3 = E_1 \cup (E_2 \cup E_3)$$

$$(E_1 \cap E_2) \cap E_3 = E_1 \cap (E_2 \cap E_3)$$

- Distributive rule

$$(E_1 \cup E_2) \cap E_3 = (E_1 \cap E_3) \cup (E_2 \cap E_3)$$

$$(E_1 \cap E_2) \cup E_3 = (E_1 \cup E_3) \cap (E_2 \cup E_3)$$

Set Theory: De Morgan's Rule

Rule-1

$$\overline{E_1 \cup E_2} = \overline{E_1} \cap \overline{E_2}$$

← Prove using the Venn Diagrams

Linguistics

opposite (Either Or) = Neither-Nor

Rule-2

$$\overline{E_1 \cap E_2} = \overline{E_1} \cup \overline{E_2}$$

← Prove using Rule-1

Probability rules

Notation: $P_2(AB) = P_2(A \cap B)$

- Union: $\Pr(E_1 \cup E_2) = \Pr(E_1) + \Pr(E_2) - \Pr(E_1 E_2)$

- Complement: $\Pr(\bar{E}) = 1 - \Pr(E)$

- Inclusion-exclusion:

$$\Pr(E_1 \cup E_2 \cup E_3) = \Pr(E_1) + \Pr(E_2) + \Pr(E_3) - \Pr(E_1 E_2) - \Pr(E_2 E_3) - \Pr(E_3 E_1) + \Pr(E_1 E_2 E_3)$$

$$\Pr\left(\bigcup_{i=1}^n E_i\right) = \sum_{i=1}^n \Pr(E_i) - \sum_{i < j} \Pr(E_i E_j) + \sum_{i < j < k} \Pr(E_i E_j E_k) - \dots + (-1)^{n+1} \Pr(E_1 E_2 \dots E_n)$$

Probability rules: conditional prob.

- Conditional probability: $\Pr(E_1|E_2) =$
prob of E_1 given that E_2 has occurred

$$\frac{\Pr(E_1 \cap E_2)}{\Pr(E_2)}$$

- Multiplication rule:

$$\Pr(E_1, E_2) = \Pr(E_1 | E_2) \cdot \Pr(E_2)$$

- Two events are called statistically independent, if
"Conditional prob of E_1 given E_2 has occurred"
= "Marginal prob. of E_1 "

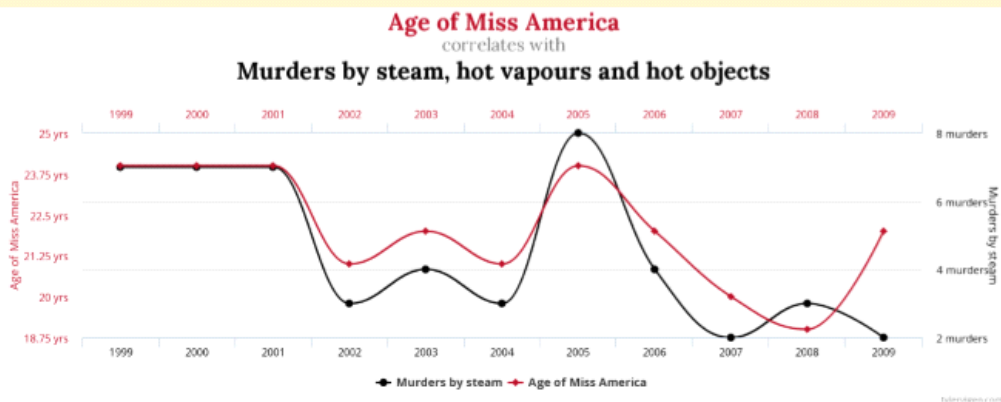
For statistically independent events

$$\Pr(E_1, E_2) = \Pr(E_1) \cdot \Pr(E_2)$$

Statistical Independence

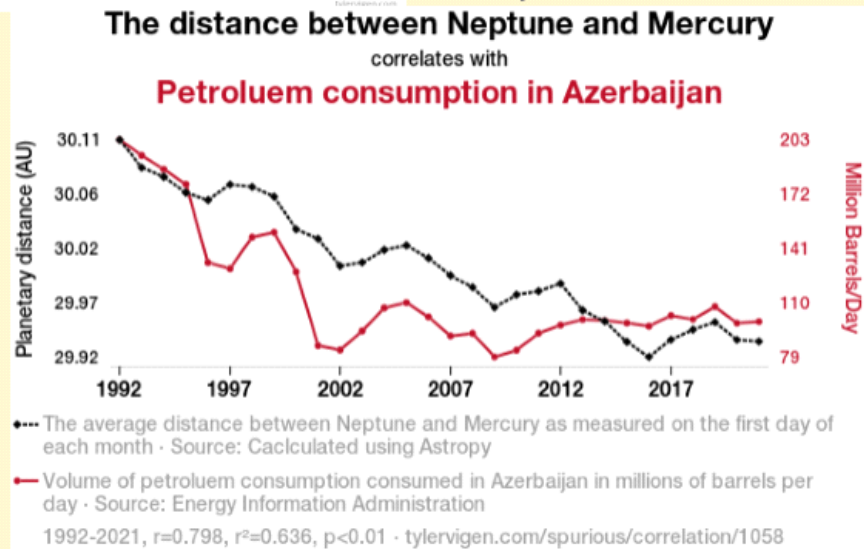
- Two events are called statistically independent, iff

Spurious Correlation



https://www.tylervigen.com/spurious/correlation/1058_the-distance-between-neptune-and-mercury_correlates-with_petroleum-consumption-in-azerbaijan

<https://www.tylervigen.com/spurious-correlations>



Probability rules: multiplication rule

$$P(E_1, E_2) = P(E_1 | E_2) \cdot P(E_2)$$

For n events,

$$P(E_1, E_2, \dots, E_n) =$$

CONDITIONALS MARGINAL

$$P(E_1 | E_2, E_3, \dots, E_n) \cdot P(E_2 | E_3, E_4, \dots, E_n) \cdots P(E_{n-1} | E_n) \cdot P(E_n)$$

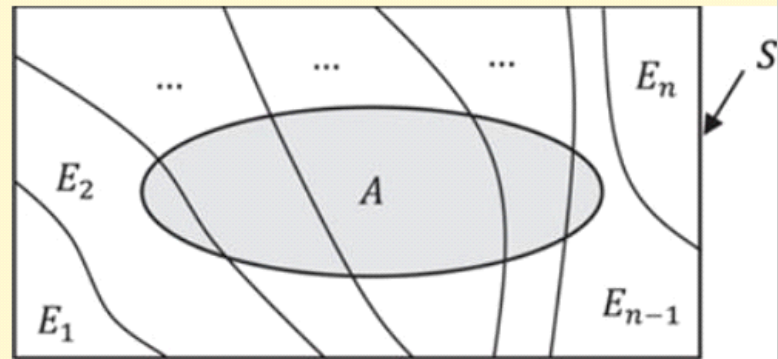
If all E_i 's are S.I.

$$P(E_1, E_2, \dots, E_n) = P(E_1) P(E_2) \cdots P(E_n)$$

Probability rules: total probability rule

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- If $E_1, E_2, \dots, E_n$ are n MECE events,



Probability rules: total probability rule

The probability that a building will sustain damage due to an earthquake depends on the intensity of shaking. Suppose the intensity is discretized into the states of weak, moderate, and strong. Assume it is determined that the probability of damage to the building is 0.01, 0.10, and 0.60 for the three levels of shaking intensity, respectively.

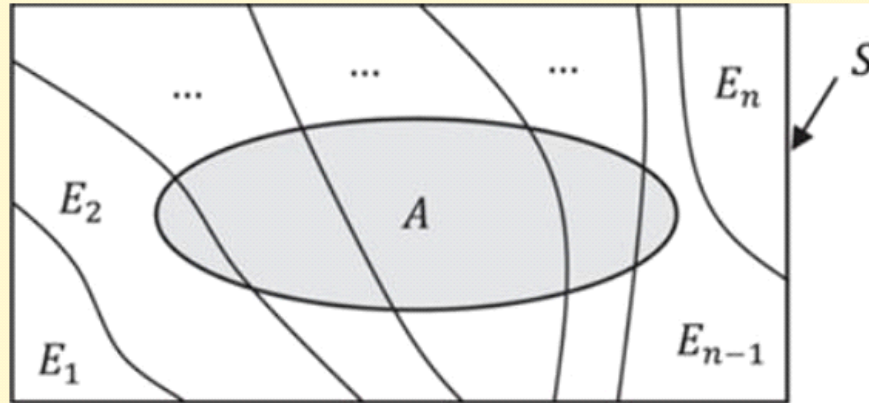
Furthermore, based on the history of past earthquakes in the region, it is estimated that, for a randomly occurring earthquake, the probabilities for the three levels of shaking intensity at the site of the building are 0.90, 0.08, and 0.02, respectively. We are interested in the probability that the building will sustain damage during the next earthquake in the region.

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Probability rules: Bayes' rule

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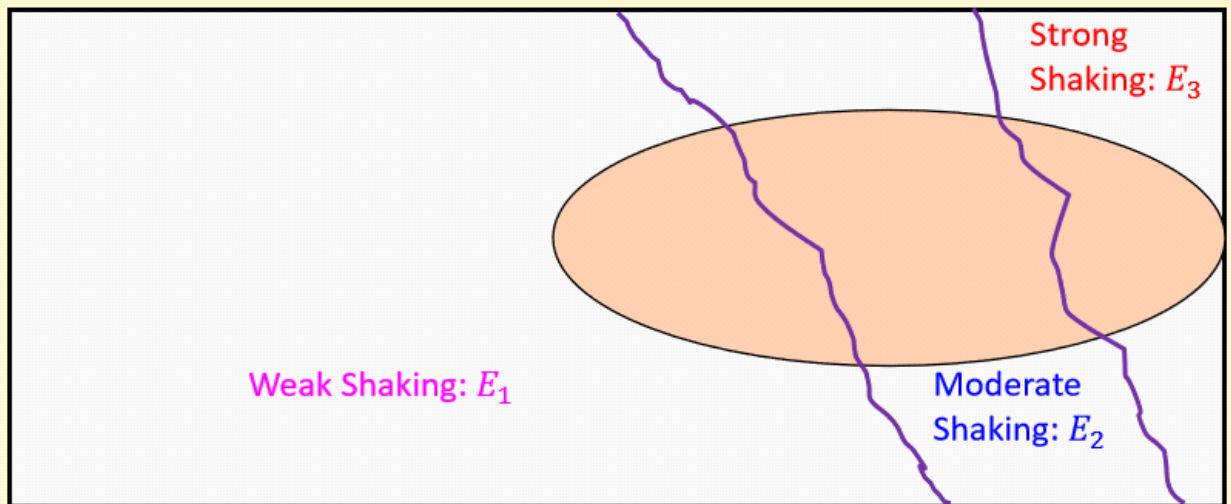
- If $E_1, E_2, \dots, E_n$ are n MECE events,



Probability rules: Bayes' rule

- Assume in the previous example that the building has sustained damage. What can we learn about the intensity of the earthquake?

Bayes' rule: Behind the Scenes!



Tutorial-1

- Topics
 - Probability
 - Venn diagrams
 - Conditional probability
- Questions: ~10-12 (question print-outs on the same day)
- Write your name, roll numbers on each sheet; bring staplers
- Closed-phone, closed-tablet, closed-AI
- Open-book, open-friends, open-TAs
- BYO
 - Empty/sheets, pen/pencil, calculators

Thank you!